# Can a Machine Think? KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

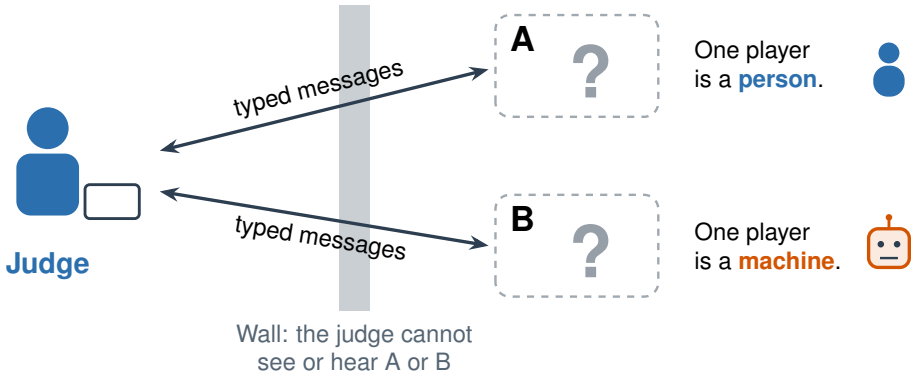
**Word bank:** judge machine passes person typed

## 1. Is it thinking? Check one box for each.

| | Yes | No | Not sure |
|---|---|---|---|
| A calculator adds 375 + 498. | ☐ | ☐ | ☐ |
| A dog finds its way home. | ☐ | ☐ | ☐ |
| A phone opens when it sees your face. | ☐ | ☐ | ☐ |
| You solve a puzzle. | ☐ | ☐ | ☐ |
| A plant turns toward the sun. | ☐ | ☐ | ☐ |
| A chatbot writes a poem. | ☐ | ☐ | ☐ |

Answers vary. People don't agree on what "thinking" means. That's the point.

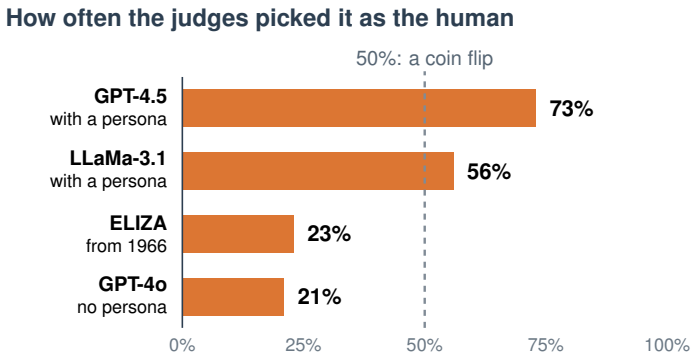
## 2. The imitation game (the Turing test)




A judge asks questions. One player is a person, and one is a machine.

The judge only reads typed answers. If the judge can't tell, the machine passes.

## 3. Has a machine passed? (a test from 2025)




Which chatbot was picked as the human most often? GPT-4.5 (73% of the time)

What did the scientists tell it to pretend to be? a shy young person who uses slang

**4. Play the imitation game.** Work in a group of 3 or 4: judge, human player, AI helper (and scorekeeper).

| Round | The judge's question | The judge **picked** (A or B) | **Was the judge right?** (yes / no) |
|---|---|---|---|
| 1 | Answers vary. | | |
| 2 | | | |
| 3 | | | |
| 4 | | | |
| 5 | | | |

The judge was right _____ times.

Our best question for catching the AI: Answers vary. Example: What is the weather now?

**5. Talk about it.** If a machine fools you, does that mean it thinks?

If a machine fools me, it **does** / **does not** mean the machine thinks, because Answers vary.

**Go Further (Day 2).** The persona experiment. The judge was right . . .

with **no** persona: _____ out of 5 with the persona: _____ out of 5 Harder to catch: Answers vary.

✂ *Cut out the cards. Add them to the class timeline.*

## 1950

Alan Turing asks: **Can a machine think?** He invents the imitation game (the Turing test).

Draw a picture here.

## 2025

In a Turing test, judges pick a chatbot as the human **73%** of the time. It used a persona.

Draw a picture here.